

How much storage does a renewable system need?

Storage and flexibility

Reading. Zerrahn et al. (2018), with the lecture note *Energy System Models* (ESM below), section 4, as the toolbox. Optional: Butters et al. (2025). Section, figure and equation numbers are the paper's own. Most questions are answered from the paper; those marked *Perspective* set it against ESM or against the evidence printed here.

Part 1 · Reading Zerrahn et al.

35 min

Why storage needs are two orders of magnitude below Sinn's

Zerrahn et al. (2018) reply to Sinn's claim that integrating wind and solar needs storage of the order of the surplus energy they produce. Sections 2–4 argue with data and a spreadsheet, section 5 with a cost-minimising model (part 2). Their one analytical device is the *residual load duration curve* (§3.4). Figure 1 is a stylised one: residual load falls linearly from L at hour 0 to zero at hour $T - H$, and over the last H hours a surplus grows linearly to S MW. The storage has efficiencies η_c for charging and η_d for discharging, as in ESM.

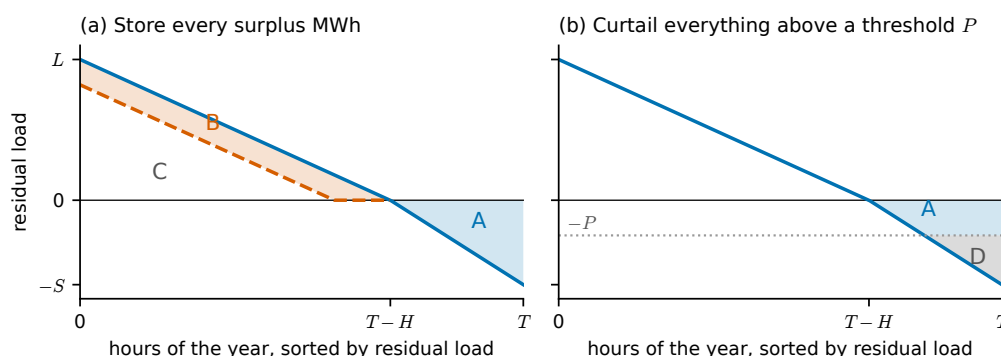


Figure 1: A stylised residual load duration curve in the paper's layout (its Fig. 4 and 5). (a) Sinn's rule: every surplus MWh (area A) is stored and released into hours of positive residual load (area B); area C is left to conventional plants. (b) Power-oriented curtailment: everything above a threshold P (area D) is thrown away.

1. The claim (§1–2).

- (i) What does Sinn claim, and which two assumptions do the authors alter?
- (ii) What does Fig. 1 show, and where is Sinn in it?

2. Sinn's rule and the duration curve (§3).

- (i) State the storage heuristic of §3.2 and explain Figure 1(a) with it: the areas, and how much energy passes through the tank in a year.
- (ii) How does the duration curve treat time, and what does that mean for the tank size?

Hint: compare one ten-hour surplus with ten one-hour surpluses, each followed by an hour of deficit. Area A is the same; is the tank?

3. Two curtailment strategies (§4).

- (i) What distinguishes the two strategies, and what does each target?
- (ii) What do they do to the tank at 50% renewables and 5% curtailment? Use question 2 to say why the second works so much better.
- (iii) On Figure 1(b), show that a threshold $P \leq S$ curtails the share $q = (1 - P/S)^2$ of the surplus energy, invert it, and find P for 5% and 20%. What does the slope at $q = 0$ say?
Hint: area D is a triangle similar to area A.
- (iv) Now read the condition for the efficient optimum at the end of §2. Why does your answer to (iii) make some curtailment efficient rather than wasteful?

4. Perspective: foresight. The storage of ESM §4.1 holds energy for the best hour ahead; Sinn's releases it at the first positive residual load.

- (i) Into which hours of the duration curve does each release? Footnote 16 says how Figure 1(a) comes to draw B as it does.
- (ii) What does foresight buy the planner, and where does the paper put market reality (footnote 29)?

Stop point

The two assumptions; $\frac{1}{2}HS$ through the tank and $\eta_c \eta_d$ of it in B; the tank is set by the longest surplus run, not by area A; $P = S(1 - \sqrt{q})$, so $0.78 S$ for 5% and $0.55 S$ for 20%.

If you finish early

Three hours. Hour 1 is windy: load 3 GW, wind 3.5 GW. Hours 2 and 3 are calm with loads 1.5 and 2.5 GW; coal has 2 GW at 40 EUR/MWh and gas 2 GW at 80. A small battery (rating $\bar{s}$, two hours of tank, $\eta_c = 0.9$, $\eta_d = 1$) ends where it started. What does Sinn's rule earn per MW, what does the optimum earn, and what is θ_h at the optimum? Which of the paper's areas B_1, B_2, B_3 (§5.2) does the optimum reach that the rule does not?

The cost-minimising model

Section 5 of the paper replaces Sinn's arithmetic with a planner: a linear programme, equations (1)–(5), that chooses renewable, conventional and storage capacities together with their hourly dispatch, so as to serve a year of demand at least cost subject to a minimum renewable share. This part reads that section. The analysis is done with ESM's tools: multipliers on the constraints, and the first-order conditions they give. We keep the paper's i for annualised investment costs, v for variable costs and C_h for curtailment, and write S_h^{in} and S_h^{out} for its charging and discharging flows, η_c and η_d for its efficiencies, and α for its minimum renewable share (the paper's φ). The multipliers have ESM's names: λ_h on the balance (2), $\bar{\mu}_{c,h}$ on the capacity bound (3a), v_h on the renewable balance (3b), θ_h on the storage balance (4a), $\bar{\phi}_h^c$ and $\bar{\phi}_h^d$ on the rating bounds (4b)–(4c), $\bar{\psi}_h$ on the energy bound (4d), and σ on the share constraint (5). The

Lagrangian of the planner's problem is

$$\begin{aligned}
L = & i^{se} N^{se} + i^{sp} N^{sp} + \sum_c i^c N^c + i^r N^r + \sum_{c,h} v^c G_h^c + \sum_h v^s (S_h^{in} + S_h^{out}) \\
& + \sum_h \lambda_h (d_h - \sum_c G_h^c - G_h^r - S_h^{out}) + \sum_{c,h} \bar{\mu}_{c,h} (G_h^c - N^c) + \sum_h v_h (G_h^r + S_h^{in} + C_h - \gamma_h N^r) \\
& + \sum_h \theta_h (S_h - S_{h-1} - \eta_c S_h^{in} + S_h^{out} / \eta_d) + \sum_h \bar{\phi}_h^c (S_h^{in} - N^{sp}) + \sum_h \bar{\phi}_h^d (S_h^{out} - N^{sp}) + \sum_h \bar{\psi}_h (S_h - N^{se}) \\
& + \sigma \left(\sum_{c,h} G_h^c - (1 - \alpha) \sum_h d_h \right),
\end{aligned}$$

with the multipliers on inequalities nonnegative and every variable nonnegative. Where a variable is strictly positive at the optimum, its first-order condition holds with equality:

$$\begin{aligned}
N^{sp} : i^{sp} &= \sum_h (\bar{\phi}_h^c + \bar{\phi}_h^d), & N^{se} : i^{se} &= \sum_h \bar{\psi}_h, \\
N^c : i^c &= \sum_h \bar{\mu}_{c,h}, & N^r : i^r &= \sum_h \gamma_h v_h, \\
G_h^c : \lambda_h &= v^c + \sigma + \bar{\mu}_{c,h}, & G_h^r : \lambda_h &= v_h, \\
S_h^{in} : \eta_c \theta_h &= v_h + v^s + \bar{\phi}_h^c, & S_h^{out} : \lambda_h &= \theta_h / \eta_d + v^s + \bar{\phi}_h^d, \\
S_h : \theta_{h+1} &= \theta_h + \bar{\psi}_h, & C_h : v_h &\geq 0, \text{ with } v_h = 0 \text{ whenever } C_h > 0.
\end{aligned}$$

1. The planner's problem (§5.1).

- (i) Explain the planner's problem in your own words: what is minimised, what is chosen, and what each of (2)–(5) requires. Which constraint does each multiplier price, and which multipliers must be nonnegative?
- (ii) Explain the four capacity conditions and the condition for G_h^c in words: which hours pay for a MW of pumps, which for a MWh of basin, which for a MW of turbines, and what σ does to a conventional plant. Why does the planner build turbines whose output is sometimes worth nothing?
Hint: use the conditions for G_h^r , S_h^{in} and C_h to say what v_h is in an hour when renewables serve load, when they fill the tank, and when they are curtailed.
- (iii) Put the condition for N^{se} together with the surplus of part 1, which arrives in a few long runs. Why does the planner stop at a tank far below Sinn's, and why is it content to curtail?
- (iv) *Perspective:* the remaining conditions, for S_h^{in} , S_h^{out} and S_h , are those of the storage model of ESM §4.1. How does the planner's problem differ from that model, and which of ESM's results carry over unchanged?

2. The duration curve at the optimum (§5.2). The paper reads its 80% solution off Fig. 10.

- (i) In the paper's words, what are the areas D_1 , D_2 , B_1 , B_2 and B_3 ?
- (ii) What are the two "economic values of electrical storage" the paper names, and which areas carry each?

3. The numbers (§5.3).

- (i) What does the paper find for Germany at 50% and at 90%: storage energy, storage power, the energy-to-power ratio, and what does it conclude about long-term storage?
- (ii) Why does the energy-to-power ratio rise with the renewable share, and why does optimal curtailment rise with it?

Hint: which hours pay for a MW of pumps, and which for a MWh of basin? How does each set of hours change as the fleet grows?

4. Power-to-x (§5.4). The paper's last experiment adds a flexible new load.

- (i) What is added to the model, and what happens to storage and curtailment at 50% and at 90% (Fig. 12)?
- (ii) In one sentence: what does a shiftable load share with a storage, and what does it lack?

Stop point

The first-order conditions in words: a MW of pumps is paid for by the hours it runs at its rating, a MWh of basin by the hours the tank is full, a MW of turbines by the hours its output is not curtailed, and σ is a tax on conventional energy. A skewed surplus fills the last MWh of basin too rarely to pay for it. Germany: 35 GWh and 6 GW at 50%, no long-term storage, and 35 to 4 GWh with power-to-x.

If you finish early

Appendix C re-runs the model with lithium-ion costs: cheap in MW, costly in MWh, and a higher round-trip efficiency. Use the first-order conditions to predict what happens to N^{se} , N^{sp} , the energy-to-power ratio and curtailment, then check against the appendix. What does the change say about the paper's §6 point that storage is a family of technologies rather than one?

What the paper's model says about Denmark

The paper runs its planner on Germany's 2014 data. We have run the same model, equation for equation, on DK1's load and weather for 2024 (24.1 TWh of demand, a peak of 4.2 GW), with the renewable share target swept from 50 to 95%. Table 1 reports the solutions, Figure 2 places them on the axes of the paper's Fig. 1, and Figure 3 draws the 80% solution as a residual load duration curve, the paper's Fig. 10 for DK1. From part 2 you need one result: σ , the multiplier on the share constraint, enters a conventional plant's condition as a tax of σ per MWh of conventional energy.

How the model is set up. Costs are ESM's tables for 2050, annuitised. There is one renewable technology, in DK1's own mix of 48% onshore wind, 21% offshore wind and 31% solar, with a capacity factor of 28% and an annuity of 89 thousand EUR per MW and year. Two gas plants play the paper's base and peak roles: a combined-cycle turbine (CCGT, annuity 100 thousand EUR per MW, 63 EUR/MWh to run) and an open-cycle one (OCGT, 45 thousand, 86 EUR/MWh). Storage is ESM's lithium-ion battery with power and energy priced separately, 9.7 thousand EUR per MW of inverter and 6.4 thousand per MWh of cells a year, which is the paper's Appendix C case. As in the paper, curtailment is free, the storage charges only from renewables, there is no carbon price, and the system is an island.

Three variations. The lower blocks of Table 1 change one thing each. *With imports* adds ESM's 5.5 GW import connection at the neighbours' 2024 hourly prices; imports count as conventional energy in the share constraint (5). *Long* prices the cells at a tenth of their cost, a stand-in for a long-duration technology. *Free* lets the storage charge from any source, not only from renewables. Question 1 uses the second, question 4 the first and the third.

1. DK1 against the paper's conclusions.

target	achieved	N^r , GW	conv., GW	N^{se} , GWh	N^{sp} , GW	E/P, h	curtailed %	σ , €/MWh
<i>The paper's model on DK1, islanded</i>								
50%	68%	7.3	3.1	3.91	0.48	8.1	7.5	0.0
70%	70%	7.6	3.0	5.15	0.60	8.6	8.8	5.1
80%	80%	9.4	2.7	15.97	1.43	11.2	14.9	41.9
90%	90%	12.1	2.2	37.77	2.36	16.0	25.6	145.2
95%	95%	13.0	1.8	86.93	2.79	31.2	26.6	458.5
<i>With the 5.5 GW import backstop, imports counted as conventional energy</i>								
50%	61%	6.5	0.0	10.64	2.12	5.0	4.1	0.0
70%	70%	7.6	0.0	12.07	2.20	5.5	7.2	13.4
80%	80%	9.2	0.0	20.33	2.47	8.2	13.1	51.0
90%	90%	12.0	0.0	39.96	2.84	14.1	25.0	167.9
95%	95%	13.0	0.0	87.34	2.99	29.2	26.6	477.2
<i>Two sensitivities, islanded: cells at a tenth of the cost (long), and charging from any source (free)</i>								
80%, long	94%	10.0	1.5	189.01	2.77	68.2	5.1	0.0
90%, long	94%	10.0	1.5	189.01	2.77	68.2	5.1	0.0
80%, free	80%	9.4	2.7	15.97	1.43	11.2	14.9	41.9
90%, free	90%	12.1	2.2	37.77	2.36	16.0	25.6	145.2

Table 1: The paper's model on DK1. "Achieved" is the renewable share the solution actually has; σ is the multiplier on the share constraint (5), in EUR per MWh of conventional energy; curtailment is a share of the renewable energy the weather offered.

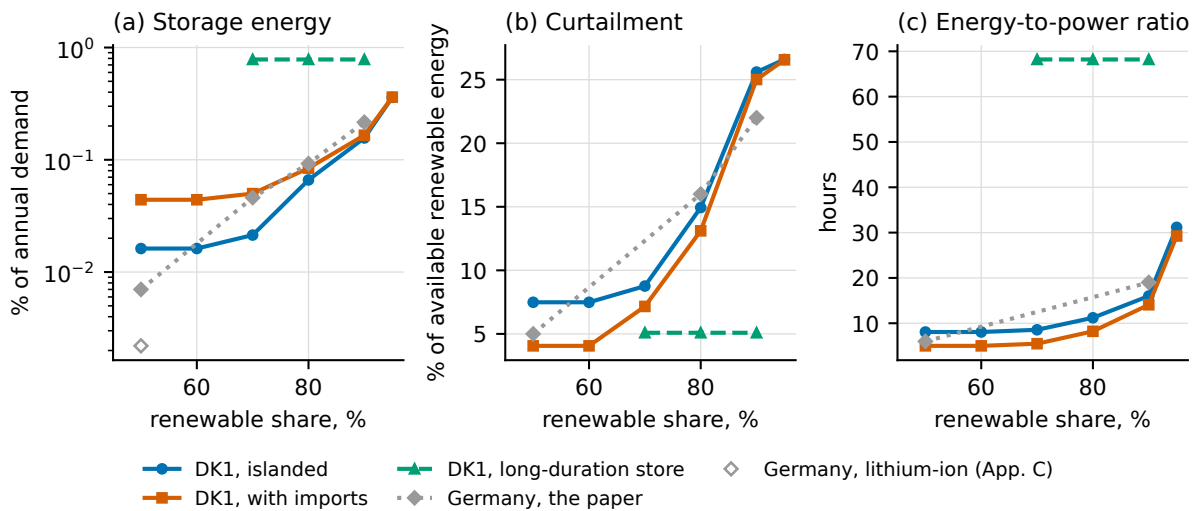


Figure 2: DK1 against the paper's Germany: (a) storage energy as a share of annual demand, the axis of the paper's Fig. 1; (b) curtailment as a share of the renewable energy available; (c) the energy-to-power ratio of the storage. Germany's points are the paper's numbers.

- Read DK1's storage at 80% from Table 1. Express it in percent of annual demand and in hours of DK1's average load, and place it on Figure 2(a) next to Germany (§5.3) and next to Sinn's numbers (§2).
- Set DK1's curtailment at 80% against Germany's optimal curtailment in §5.3.
- The "long" rows price the cells at a tenth of their cost. What energy-to-power ratio does the planner then choose, and is that the "true long-term storage" §5.3 says is never needed?

2. The duration curve for DK1.

- The dashed curve in Figure 3 has three flat parts. Explain what sets each. The lowest is minus the storage's power rating; check it against the N^{sp} column of Table 1 at 80%.

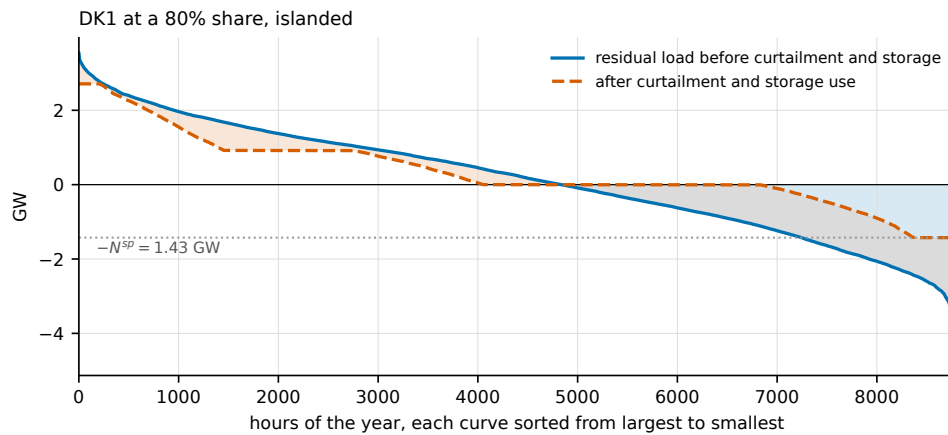


Figure 3: DK1's residual load duration curve at an 80% share, islanded, drawn as the paper's Fig. 10: before curtailment and storage, and after both, each sorted on its own. The dotted line is minus the storage's power rating.

- (ii) The figure is the paper's Fig. 10 for DK1 (§5.2, part 2's question 2). Mark the five areas D_1 , D_2 , B_1 , B_2 and B_3 on it.

3. What the target costs.

- (i) σ is zero at a 50% target. What does that mean, and what share does the planner reach on cost alone? Use the costs in the setup to say why it is there.
Hint: divide the renewable annuity by the MWh a MW delivers in a year.
- (ii) Read σ at 80% from Table 1 and convert it to a carbon price, using that a CCGT burns 1/0.6 MWh of gas per MWh with 0.198 t of CO_2 per MWh of gas.

4. Perspective: §6 tested on DK1. Table 1 tests two things the paper left out, and in both DK1 disagrees with it: interconnection (§6, fifth point), which the paper expects to lower storage needs, and the remark at the end of §5.2 that charging from conventional plants "would strengthen both economic values".

- (i) With imports, storage *rises*: 20.33 GWh against 15.97 at 80%. Footnote 17 says what interconnection does to the surplus. What can a one-way import connection do for area A, and what can it not?
- (ii) The "free" rows are identical to the islanded ones. Why does charging from the gas plants never pay here? The two plants in the setup above, their running costs, and σ are what you need.

Stop point

DK1's storage at 80% in GWh, in percent of annual demand and in hours of average load, against Germany's; the share the planner reaches unforced; and σ at 80% as a carbon price.

If you finish early

Apply Sinn's 3.23% at 89% to DK1 and set it against the table's 90% row. Then Appendix B: the weather year changes the optimal storage by a factor of five. Name the statistic of the hourly residual load that would predict which of DK1's years 2019 to 2024 needs the most storage, and say why the duration curve cannot provide it.